

PS No:	M	O	D	E	L			Student Number:	S	O	L	U	T	I	O	N	
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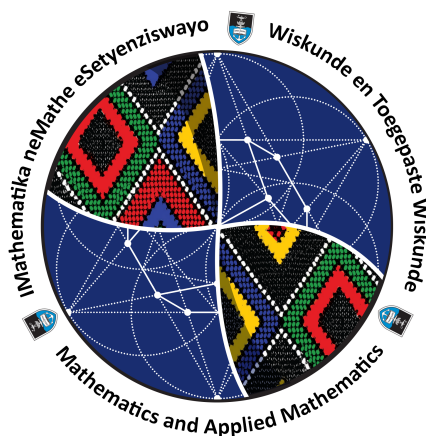
Department of Mathematics and Applied Mathematics
MAM3000W 3DM: Discrete Mathematics

Test 1

22 March 2024

Time : 45 minutes

40 marks = 100%



- This question paper consists of **8 pages** (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. The work in your *UCT Examination Answer Book* will not be marked. Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.
- For Question A, write the answer only. If you show working you will be penalized.
- For Question B you must prove everything. No proof = no marks, even if your answer is correct.

Question A. [25 points]

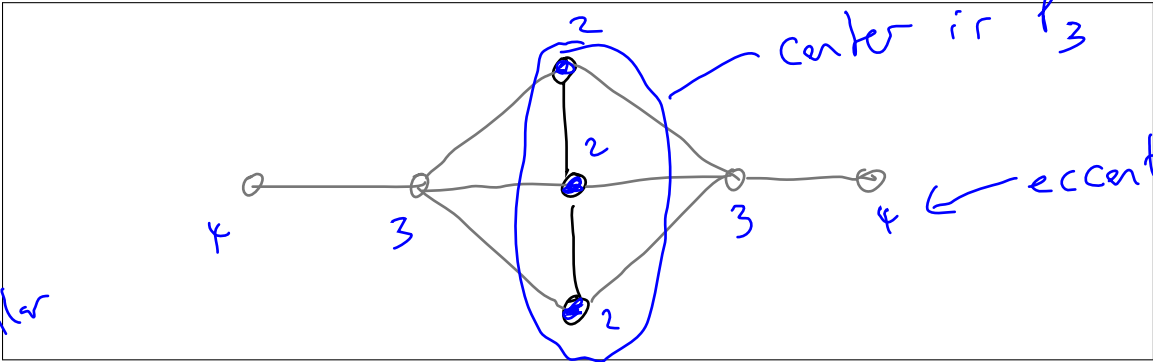
1. Draw an example of each of the following (if such a graph exists) or write Does Not Exist (if it doesn't). No proofs or working necessary.

- (a) A graph with degree sequence 5, 4, 3, 2, 1, 0. (3)

Does Not Exist

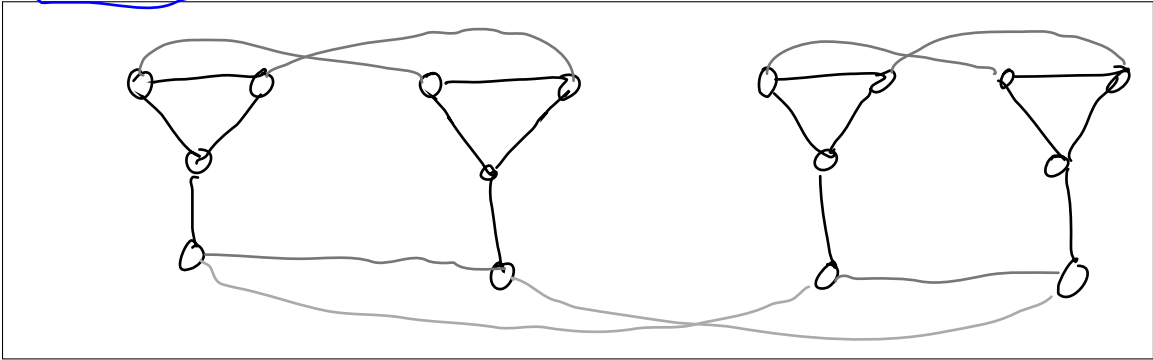
The degree sequence has an odd number of odd numbers.

- (b) A graph with center P_3 . (3)



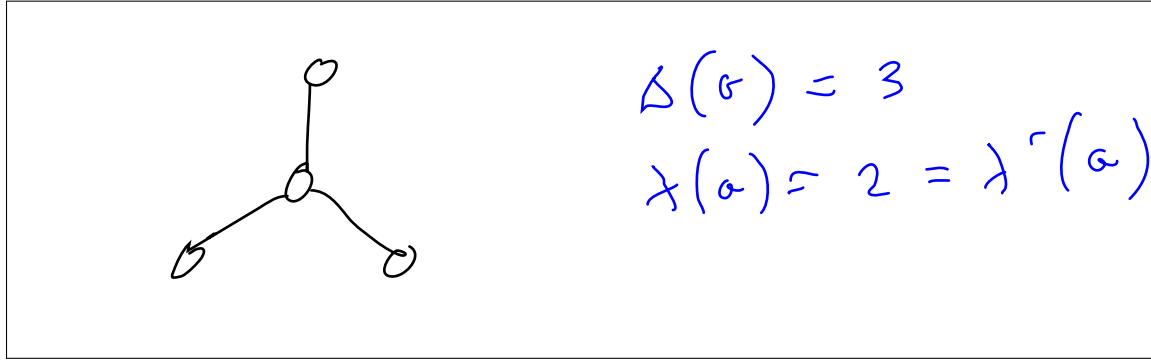
3-regular

- (c) A cubic graph that contains the martini glass as an induced subgraph. (3)



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- (d) A graph G with $\Delta(G) > 2$ and $\lambda(G) = \lambda^-(G)$. (3)

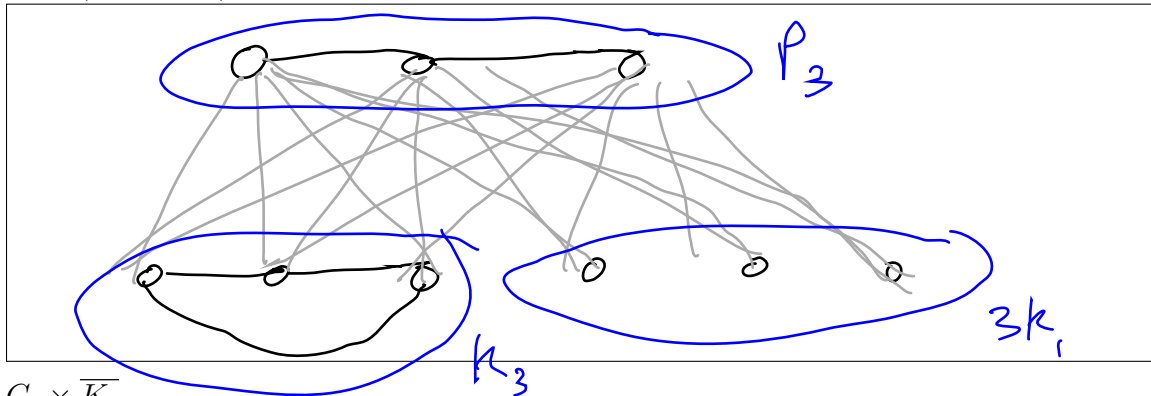


- (e) A digraph of order 9 that is strongly connected but not weakly connected. (3)

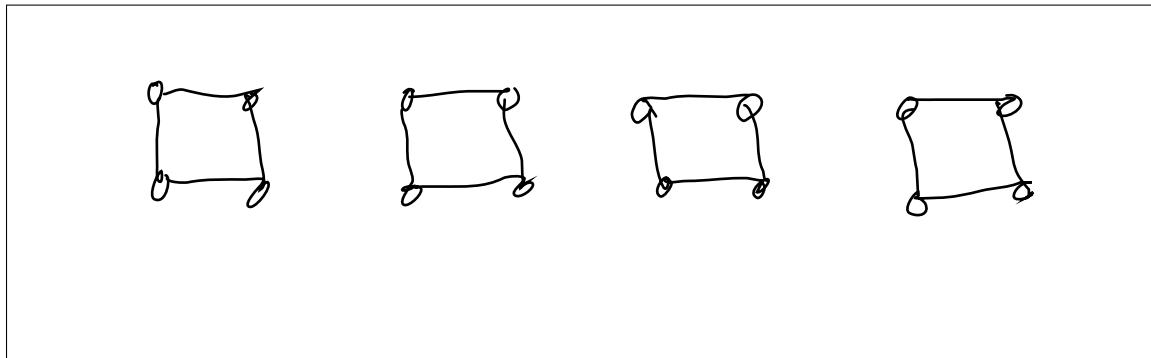
Does Not Exist
 Strongly connected $\Rightarrow$ weakly connected

2. Draw the following graphs:

- (a) $P_3 + (K_3 \cup 3K_1)$. (3)



- (b) $C_4 \times \overline{K_4}$. (3)



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3. Define the following terms:

(a) Connected graph.

(2)

A graph G is connected if for all $u, v \in V(G)$, there exists a $u-v$ path.

(b) Automorphism of a graph.

(2)

An automorphism of a graph G is a permutation α of $V(G)$ such that $uv \in E(G) \Leftrightarrow \alpha(u)\alpha(v) \in E(G)$.

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Question B. [20 points]

For Question B, show all working/proofs! Zero working/proofs = zero marks!

1. A graph G is *irregular* if no two vertices of G have the same degree. Prove that if G is an irregular graph, then G is trivial. (5)

Suppose, to the contrary, that there exists a nontrivial irregular graph G of order n , and note that $n \geq 2$. Every vertex v of G has $\deg v \in [0, n-1]$. Since there are n vertices and $|[0, n-1]| = n$, there must be a vertex x with $\deg x = 0$ and a vertex y with $\deg y = n-1$ (and since $n \geq 2$, $x \neq y$). But this is not possible: the vertex x is not adjacent to y , but the vertex y is adjacent to x . Thus no such graph exists.

QED

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2. If G is a connected graph, prove that

$$\text{diam } G \leq 2 \text{ rad } G.$$

(5)

let $u, v, c \in V(G)$ such that $d(u, v) = \text{diam } G$
 and $e(c) = \text{rad } G$. Then $d(u, c) \leq \text{rad } G$
 and $d(c, v) \leq \text{rad } G$, so

$$\begin{aligned} \text{diam } G = d(u, v) &\leq d(u, c) + d(c, v) \\ &\quad (\triangle \text{-inequality}) \\ &\leq \text{rad } G + \text{rad } G, \end{aligned}$$

$$\text{ie. } \text{diam } G \leq 2 \text{ rad } G.$$

QED

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3. If G is a disconnected graph, prove that

$$\text{diam } \bar{G} \leq 2.$$

(5)

Let $u, v \in V(G)$, $u \neq v$. Note that $V(G) = V(\bar{G})$.

We consider two cases:

(1) $uv \notin E(G)$. Then $uv \in E(\bar{G})$, so
 $d_{\bar{G}}(u, v) = 1$.

(2) $uv \in E(G)$. Then u, v are in the same component C of G . Since G is disconnected, it has a component $C' \neq C$. Let $z \in V(C')$. Then
 $uz \notin E(G)$ and $vz \notin E(G)$, so
 $uz \in E(\bar{G})$ and $vz \in E(\bar{G})$,
and thus $d_{\bar{G}}(u, v) = 2$.

Hence $\text{diam } \bar{G} \leq 2$.

QED

4. Prove or disprove: For every positive integer t , the hypercube Q_t is bipartite.

(5)

This is True. Recall that Q_t is the graph where $V(Q_t) = \{b : b \text{ is a binary string of length } t\}$ and $bb' \in E(Q_t) \Leftrightarrow b \text{ and } b' \text{ differ in exactly one digit.}$

Proof 1: let $V_1 = \{b \in V(Q_t) : b \text{ has an even number of 1's}\}$
 $V_2 = \{b \in V(Q_t) : b \text{ has an odd number of 1's}\}.$

Clearly, this is a partition of $V(Q_t)$.

Furthermore from the definition of Q_t , no two vertices of V_1 are adjacent and no two vertices of V_2 are adjacent, so $\{V_1, V_2\}$ is a bipartition of $Q_t \Rightarrow$ thus bipartite. (QED)

Proof 2: let C be a cycle of Q_t and $u \in V(C)$.
 As we move around C from u , every edge corresponds to one digit changing. Every digit of u must change an even number of times, so C is even, (QED)
 so Q_t is bipartite.